

SQL Query Execution & Verification Report

ONLINE BOOKSTORE ANALYTICS PROJECT

Database: MySQL Workbench
Author: Hafiz Rehman

QUERY #1

Total Books Sold by Genre

Logic: Aggregates order volumes grouped by genre using a JOIN between Orders and Books.
Finding: *Mystery* leads volume with 504 units sold, followed by *Science Fiction* (447) and *Fantasy* (446).

```
00 -- 1) Retrieve the total number of books sold for each genre
01 SELECT b.Genre, SUM(o.Quantity) AS Total_Books_Sold
02 FROM Orders o
03 JOIN Books b ON o.Book_ID = b.Book_ID
04 GROUP BY b.Genre;
05
```

Genre	Total_Books_Sold
Biography	285
Fantasy	446
Science Fiction	447
Mystery	504
Romance	439
Non-Fiction	351
Fiction	225

QUERY #2

Average Price of "Fantasy" Genre

Logic: Calculates the scalar mean price for all inventory titles under Fantasy using AVG().
Finding: The average retail catalog price across Fantasy titles evaluates to exactly **\$25.98**.

```
106 -- 2) Find the average price of books in the "Fantasy" genre
107 SELECT AVG(Price) AS Average_Price
108 FROM Books
109 WHERE Genre = 'Fantasy';
```

Average_Price
25.981690

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QUERY #3

Customers With At Least 2 Orders

Logic: Groups orders by Customer_ID and filters multi-order customers using HAVING COUNT () >= 2.

Finding: Exactly 139 repeat buyers were identified within the database.

```
111 -- 3) List customers who have placed at least 2 orders
112 • SELECT c.Customer_ID, c.Name, COUNT(o.Order_ID) AS Total_Orders
113 FROM Customers c
114 JOIN Orders o ON c.Customer_ID = o.Customer_ID
115 GROUP BY c.Customer_ID, c.Name
116 HAVING COUNT(o.Order_ID) >= 2;
```

Customer_ID	Name	Total_Orders
2	Crystal Clements	2
4	Stephen Vasquez	2
8	Matthew Johnson	2
13	Kristine Kim	2
14	John Wood	2
15	Vanessa Gaines	2
16	Stacey Flores	3
21	Edge Frost	2
22	Stacey Adams	3
23	Hannah Drake	2
24	Christina Mitchell	2

QUERY #4

Most Frequently Ordered Book

Logic: Ranks books by total order occurrences and limits the result set to 1.

Finding: "Implemented encompassing conglomeration" (Book ID 31) emerged as top ordered title with 4 orders.

```
118 -- 4) Find the most frequently ordered book
119 • SELECT b.Book_ID, b.Title, COUNT(o.Order_ID) AS Order_Count
120 FROM Orders o
121 JOIN Books b ON o.Book_ID = b.Book_ID
122 GROUP BY b.Book_ID, b.Title
123 ORDER BY Order_Count DESC
124 LIMIT 1;
```

Book_ID	Title	Order_Count
31	Implemented encompassing conglomeration	4

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QUERY #5

Top 3 Most Expensive Fantasy Books

Logic: Filters for Fantasy titles, orders descending by price, and retrieves top 3 records.

Finding: The premium title is *"Stand-alone content-based hub"* priced at **\$49.90**.

```
126 -- 5) Show the top 3 most expensive books of 'Fantasy' Genre
127 • SELECT *
128 FROM Books
129 WHERE Genre = 'Fantasy'
130 ORDER BY Price DESC
131 LIMIT 3;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62
228	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

QUERY #6

Total Quantity Sold by Author

Logic: Joins Books and Orders, aggregating total quantities grouped by author.

Finding: Author *Tonya Saunders* leads overall sales with 21 total units sold.

```
133 -- 6) Retrieve the total quantity of books sold by each author
134 • SELECT b.Author, SUM(o.Quantity) AS Total_Quantity_Sold
135 FROM Orders o
136 JOIN Books b ON o.Book_ID = b.Book_ID
137 GROUP BY b.Author;
```

Author	Total_Quantity_Sold
Margaret Moore	8
John Davidson	13
Christopher Fuentes	6
Marissa Smith	16
Christopher Dixon	15
Tonya Saunders	21
Larry Hunt	6
Brandon Foster	4
Michelle Bell	11
Mary French	14
Lisa Lopez	1

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QUERY #7

Customer Cities with Spend > \$30

Logic: Filters total order amounts greater than \$30 and retrieves unique city locations via SELECT DISTINCT.

Finding: Identifies key geographic customer hubs including *Lake Paul*, *North Keith*, and *Kelseyfort*.

```
139 -- 7) List the cities where customers who spent over $30 are located
140 • SELECT DISTINCT c.City
141 FROM Customers c
142 JOIN Orders o ON c.Customer_ID = o.Customer_ID
143 WHERE o.Total_Amount > 30
```

City
Lake Paul
North Keith
Kelseyfort
East David
Richardsonville
Ramosstad
Rogersborough
New Carlsbury
Ravenberg
West Anthony
North Carolyn

QUERY #8

Highest-Spending Customer

Logic: Aggregates total transaction spend per customer, sorting descending with a LIMIT 1 clause.

Finding: Kim Turner (Customer ID 457) is the top spender with **\$1,398.90** in purchases.

```
145 -- 8) Find the customer who spent the most on orders
146 • SELECT c.Customer_ID, c.Name, SUM(o.Total_Amount) AS Total_Spent
147 FROM Customers c
148 JOIN Orders o ON c.Customer_ID = o.Customer_ID
149 GROUP BY c.Customer_ID, c.Name
150 ORDER BY Total_Spent DESC
151 LIMIT 1
```

Customer_ID	Name	Total_Spent
457	Kim Turner	1398.90

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QUERY #9

Remaining Stock After Order Fulfillment

Logic: Performs a LEFT JOIN with COALESCE to subtract fulfilled quantities from initial stock.
Finding: Real-time inventory tracking updated, revealing items requiring urgent replenishment.

```
153 -- 9) Calculate the stock remaining after fulfilling all order
154 *
155 SELECT
156     b.Book_ID,
157     b.Title,
158     b.Stock AS Initial_Stock,
159     COALESCE(SUM(o.Quantity), 0) AS Total_Ordered,
160     (b.Stock - COALESCE(SUM(o.Quantity), 0)) AS Remaining_Stock
161 FROM Books b
162 LEFT JOIN Orders o ON b.Book_ID = o.Book_ID
163 GROUP BY b.Book_ID, b.Title, b.Stock;
```

Book_ID	Title	Initial_Stock	Total_Ordered	Remaining_Stock
1	Configurable modular throughput	100	3	97
2	Personalizing reciprocal knowledge user	19	0	19
3	Streamlined coherent initiative	27	5	22
4	Customizable 24-hour product	8	0	8
5	Adaptive 5th generation encoding	16	8	8
6	Advanced encompassing implementation	2	0	2
7	Open-architected existing structure	95	5	90
8	Persistent local encoding	84	3	81
9	Optimized interactive challenge	70	0	70
10	Ergonomic relational hub	25	1	24
11	Secured zero tolerance time-frame	10	5	5

EXECUTIVE TAKEAWAYS & KEY NEXT STEPS

The executed queries validate data integrity across orders, customers, and inventory levels in MySQL. Strategic recommendations include:

- Target High-Volume Genres:** Focus promotional activity on *Mystery* and *Fantasy*, which dominate total unit sales.
- VIP Retention Program:** Launch targeted rewards for the 139 high-value repeat customers to maximize retention.
- Inventory Automation:** Implement triggers for low-stock items where `Remaining_Stock < 10` to prevent fulfillment delays.